



LIFELINE MEDICAL TECHNOLOGIES
Precision Analgesia

The **Active Safety Net** for Inpatient Opioid Prescribing

A SMART on FHIR Clinical Decision Support engine that surfaces a specific, actionable, guideline-cited recommendation at the moment of prescribing — which the clinician reviews and can accept or override, with every override logged — designed to embed inside Epic and Cerner.



ENTERPRISE B2B SAAS



SELF-ASSESSED NON-DEVICE CDS



OPIOID SAFETY

CONFIDENTIAL — FOR INVESTOR REVIEW ONLY



THE INPATIENT OPIOID BLIND SPOT

Hospitals are writing opioid orders **without a safety net.**

\$78.5B

Annual U.S. economic burden of prescription opioid misuse, abuse, and dependence

Florence et al., Medical Care 2016

Litigation Risk

Opioid-dosing and monitoring failures are an active, growing category of hospital liability exposure nationally.

Illustrative framing — not a specific case citation

46%

Of monitored inpatients on continuous monitoring experienced opioid-induced respiratory depression (OIRD) events. Manual nursing checks miss the vast majority.

Khanna AK et al., Anesthesia & Analgesia, 2020;131(4):1012-1024 (PRODIGY Trial)



Alert Fatigue Is the Norm

Epic's native drug-safety warnings are dismissed **70–90% of the time (JAMIA)**. Clinicians have been trained to click past them. The system meant to protect patients has been optimized away.



Settlement Abatement Funds — Potentially Eligible Use

Point-of-care decision support is a **potentially eligible, discretionary remediation-spend category** under a hospital's Exhibit E settlement allocation — subject to that hospital's own settlement administrator and state-specific allowable-use rules. Not a designated or guaranteed funding source.



WHY EXISTING TOOLS FAIL

The difference between **reactive** and **proactive**.

STATUS QUO

The Passive Calculator

- ✗ Doctor must consciously *remember* to open MDCalc or a calculator app
- ✗ Manually types in creatinine, weight, opioid doses — 45 seconds of friction
- ✗ Used only 5% of the time. Used in the cafeteria, not at the bedside.
- ✗ Boolean alerts ("MME > 90") dismissed in under 1 second
- ✗ No pharmacogenomics, cross-tolerance, or organ-specific gating

RESULT

Optional. Ignored. Liability.

↕
VS
↕

PRECISION ANALGESIA

The Active Safety Net

- ✓ Monitors CPOE in the background via CDS Hooks
- ✓ Auto-pulls weight, labs, meds from FHIR — **zero manual entry**
- ✓ Surfaces a specific recommendation *at the moment of prescribing* — clinician reviews and accepts or overrides, with the override logged
- ✓ Returns specific, actionable guidance (not just "warning")
- ✓ CYP2D6 pharmacogenomics, OIRD scoring, renal/hepatic gating

RESULT

Automatic. Specific. Defensible.



HOW IT WORKS

A three-layer engine embedded in every opioid order.

LAYER 1 — BUILT

FHIR Ingestion

Patient context auto-loads via SMART on FHIR OAuth. Pulls age, weight, creatinine, active meds, allergies, genotype panel. **Zero manual entry.**

R4 Patient

Observation

MedicationRequest

LAYER 2 — BUILT

Clinical Pipeline

10 deterministic safety rules plus OIRD risk scoring, MME thresholds, pharmacogenomic guidance, and CDC naloxone prompts — all in <100ms.

CDS Hooks

PRODIGY

CDC 2022

LAYER 3 — SCAFFOLDED

FHIR Write-Back

1-click "Draft Order" injects safe alternative into Epic's CPOE cart. "Export to Chart" writes a billable consult note as a DocumentReference.

MedicationRequest POST

DocumentReference

The Traceability Ledger — Supporting Our Self-Assessed Criterion D Posture



Every recommendation shows the exact formula, conversion factor, and guideline citation. The clinician can independently verify the math and is not primarily relying on the software. This is designed to be legally defensible — not a black box — pending confirmatory review by regulatory counsel.

SELF-ASSESSED NON-DEVICE CDS



CLINICAL MODULES

Six tools. One workflow. Zero new logins.



Risk Assessment

OME/MME calculator, OIRD risk tier (PRODIGY-aligned), organ dose gating, naloxone prompt at ≥ 50 MME, manual data override form.



Opioid Tapering

CDC 2022 taper schedule generator. 10%/2-week protocol. One-click copy to EHR note.



OUD Consult

BUPE/methadone/naltrexone protocols. COWS, DSM-5, aberrant behavior screening. Auto-structured consult note.



Patch Conversion

Transdermal fentanyl/buprenorphine calculators with organ-adjusted dosing. Prevents the most common ICU conversion error.



Screening Suite

DAST-10, ASSIST, ORT with sex-adjusted scoring. Reads FHIR nursing flowsheets before prompting manual entry.



IV Infusion

PCA 1-hr/4-hr limit calculator. Continuous drip converter with renal-adjusted dosing. Prevents common ICU programming errors.



Manual Override Form — "Dirty Data Defense"

Clinicians correct/supplement FHIR data in real-time. 6 sections, EHR/Manual/Missing badges.



Copy-to-Note Workflow

All outputs copy as structured clinical text. Paste into Epic note. Citrix/VDI compatible.



REGULATORY & SECURITY MOATS

Designed from day one to **avoid the two graveyards.**



THE FDA OPPORTUNITY

Self-Assessed Non-Device CDS Posture

FDA's January 2026 guidance describes a "Non-Device CDS" pathway for deterministic rules a clinician can independently verify. Our Traceability Ledger is engineered to support **Criterion D** — but this is a self-assessment, pending confirmatory review by regulatory counsel, not a guaranteed or settled classification.

Potential to avoid ~\$2M + 2 years

vs. 510(k) SaMD clearance required for AI/ML competitors — if the self-assessment holds up under counsel review



THE CISO MOAT

Zero-PHI Architecture

All calculations happen in-memory, discarded after the HTTP response. **No database. No PHI at rest.** A BAA template exists and will require counsel review and execution before any PHI-adjacent pilot. A hospital CISO's biggest fear — a data breach of stored PHI — is architecturally reduced.

Designed to ship past security review faster than a PHI-storing system



THE INTEGRATION MOAT

SMART on FHIR + CDS Hooks

We embed natively via Epic's published APIs. We never ask hospitals to change their workflows. Once a validated clinical system is integrated, hospital IT will never unwind it.



THE FUNDING ANGLE

Settlement-Eligible Sales Motion

Point-of-care CDS is a **potentially eligible, discretionary Exhibit E remediation category** of National Opioid Settlement abatement funds — subject to each hospital's own settlement administrator and state rules. Not a guaranteed or designated funding source, but a conversation that can ease the capital budget objection.



TOTAL ADDRESSABLE MARKET

A massive, underserved acute care opportunity.

TAM (ILLUSTRATIVE
ESTIMATE)**\$6.8B**

U.S. hospital CDS / patient
safety software market
(2026) — modeled estimate,
needs independent sourcing

SAM (ILLUSTRATIVE
ESTIMATE)**\$1.4B**

Inpatient opioid safety &
analgesic stewardship
software segment —
modeled estimate, needs
independent sourcing

SOM (YEAR 3 MODEL)

\$87M

Modeled assumption: 350
hospital systems × \$250K
ARR — a 3-year financial-
model projection, not a
researched figure

Addressable Hospital Universe

Total U.S. acute care hospitals	6,090
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Epic-affiliated hospitals	~2,800
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Hospitals with opioid settlement exposure	~4,100
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Hospitals with a discretionary abatement allocation (illustrative estimate)	~1,200
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Why Now — Four Simultaneous Tailwinds

- ▶ FDA Jan 2026 guidance describes a non-device CDS pathway (self-assessed, pending counsel confirmation)
- ▶ Settlement abatement funds (potentially Exhibit E-eligible, discretionary) can ease budget objections
- ▶ Epic Connection Hub is an established channel for SMART on FHIR CDS partners
- ▶ CMS opioid quality metrics now tied to hospital reimbursement



COMPETITIVE LANDSCAPE

We thread the needle between every incumbent.

COMPETITOR	APPROACH	FATAL WEAKNESS	OUR WEDGE
Epic / FDB Alerts Native EHR	Boolean pop-ups on every order	70–90% dismiss rate (JAMIA). Alert fatigue is a clinical crisis.	Multi-factor engine. Specific. Override-documented.
MDCalc / UpToDate SMART calculators	Reactive reference the doctor must launch	5% usage rate. Used in the cafeteria, not the CPOE.	Proactive. Monitors CPOE in the background. Surfaces guidance at the moment of prescribing.
Bamboo / NarxCare PDMP database	Narx Score from retail pharmacy fills	Outpatient only. Useless for ICU. Black-box backlash.	Owens inpatient. Transparent math. Works where NarxCare doesn't.
AI/ML CDS Startups Predictive AI	ML overdose risk models, hundreds of EHR variables	Likely needs 510(k) SaMD clearance. \$2M+ / 2 years. CISOs wary.	Deterministic. Self-assessed non-SaMD posture. CISO-friendly design.
Precision Analgesia ✓ SMART on FHIR CDS	Proactive CDS Hook engine, FHIR auto-population, deterministic transparent math	—	Inpatient. Proactive. Self-assessed Non-Device CDS. CISO-friendly design. Settlement-eligible sales angle.



REVENUE

From an iOS beta to a \$250K/year hospital contract.

PRIMARY REVENUE

Enterprise SaaS — Health System Licensing

\$150K–\$350K

Annual contract per hospital system — priced against the liability, quality-metric, and monitoring exposure it addresses.

Implementation & onboarding	\$15–30K one-time
Base SaaS license	\$120–250K/yr
Pharmacogenomics tier	+\$50K/yr
Settlement documentation	+\$25K/yr

SECONDARY REVENUE

iOS App — Individual Clinicians

Currently free (TestFlight beta). Serves as top-of-funnel — a hospital's CMO downloads the app, uses it at home, champions the enterprise integration. Freemium → institutional subscription path.

Unit Economics (Per Hospital System)

U.S. economic burden of Rx opioid harm (Florence et al., 2016) **\$78.5B/yr**

Annual cost of our integration **\$150–350K**

Native opioid alerts dismissed today (JAMIA) — the spend we replace **70–90%**

Settlement abatement angle (discretionary) **Exhibit E-eligible**

Go-To-Market Motion

- 1 **Epic Connection Hub** → planned channel for inbound discovery across the ~2,800-hospital Epic network
- 2 **CMIO champion** path — iOS app creates awareness, targeting CMIO-requested enterprise evals
- 3 **Settlement framing** — hospital legal explores discretionary abatement funding, potentially easing the budget conversation
- 4 **Shadow Mode** — 90-day silent run proves error detection rate before go-live



WHERE WE ARE TODAY

Engine built. Ready for the track.

21,000+

Lines of production code (frontend + backend)

25

Clinical React components (6 major workflow modules)

368

Automated clinical tests passing (264 iOS validation scenarios + 104 backend pytest tests)

15

Clinical API endpoints (rate limited + JWT auth)

✅ Built & Verified

- ✓ Full SMART on FHIR R4 patient context ingestion
- ✓ 10-rule Safety Engine + OIRD scoring, OME/MME, renal/hepatic, pharmacogenomics
- ✓ WCAG 2.1 A accessibility (screen readers, keyboard nav, VDI-compliant)
- ✓ Rate limiting + JWT token validation on every clinical route
- ✓ FHIR Write-Back scaffolding (MedicationRequest + DocumentReference)
- ✓ iOS app in TestFlight beta with real clinical users

🚀 Next 30 Days (Post-Seed)

- Generate JWKS keys → register at **fhir.epic.com**
- Prove FHIR R4 on Epic Sandbox synthetic patients
- Wire FHIR Write-Back → live 1-click Naloxone co-prescribing
- AKI trending — last 3 creatinine values, not just today's
- Submit to Epic Connection Hub — activate inbound discovery channel
- Target first pilot hospital for a Shadow Mode deployment (no pilot signed yet)



FUNDRAISE

Seed Round — putting tires on the Ferrari.

RAISING

\$2.5M

Seed Round — 18-month runway to first signed hospital contracts

Use of Funds

Epic Sandbox → Production Integration	\$600K (24%)
Clinical Validation & Pilot Hospital	\$700K (28%)
Engineering (AKI, CDS Hooks, Write-Back)	\$750K (30%)
Enterprise Sales & CMIO Outreach	\$450K (18%)

18-Month Milestones

- MONTH 3

Epic Sandbox registration, JWKS keys, FHIR R4 synthetic test passing
- MONTH 6

Target: first pilot hospital signed. Target: Shadow Mode live. CDS Hooks on medication-prescribe.
- MONTH 9

Target: Shadow Mode data demonstrates error-detection rate. Target: first \$250K contract signed. Epic Connection Hub listing target.
- MONTH 18

5+ hospital contracts. Series A raise. SOC 2 Type II audit.

💡 The Settlement-Eligible Sales Insight

Where a hospital's own settlement administrator approves point-of-care CDS as a discretionary Exhibit E remediation use, the integration **may not require a separate capital budget approval** — not a guaranteed or designated funding source. Where this framing applies, we believe it can compress an 18-month enterprise sales cycle to roughly 4–6 months. Our seed capital funds our first push toward settlement-eligible contracts.

THE VERDICT

You are not buying a calculator.
You are buying a safety net for every opioid order.

Every hospital without this tool is writing opioid orders without a safety net, against a national \$78.5B/yr opioid economic burden (Florence et al., 2016) and a well-documented alert-fatigue problem. The clinical engine is built. Our regulatory posture is self-assessed as favorable, pending confirmatory review by counsel. Settlement funds are a potentially eligible, discretionary funding angle — not a guarantee.

WEBSITE

precision.lifelinemedtech.org

IOS BETA

[TestFlight](#)

COMPANY

Lifeline Medical Technologies

